

Heterogeneous Freezing in Super-Droplet Monte Carlo Cloud μ -Physics Models

S. Arabas¹, J.H. Curtis², I. Silber³, A. Fridlind⁴, D.A. Knopf⁵, M. West² & N. Riemer²

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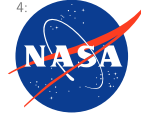
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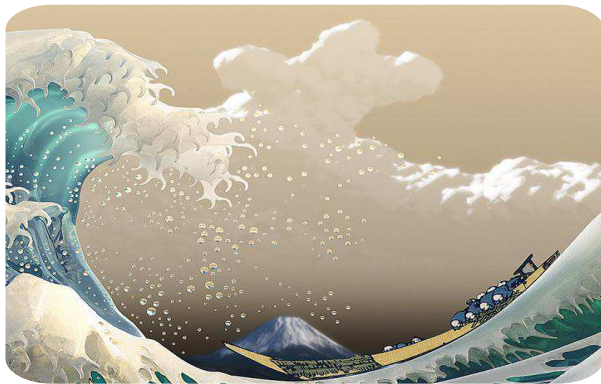
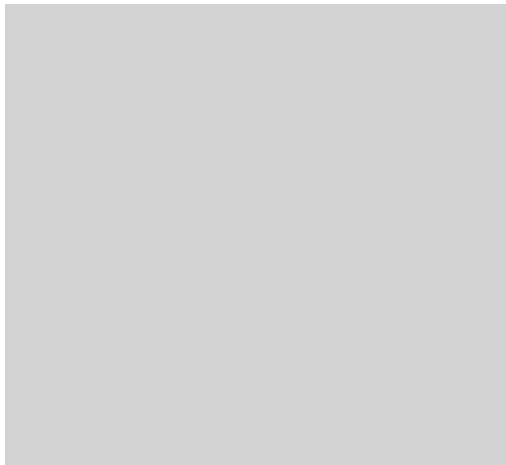
Aerosol-cloud interactions: a conceptual picture

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background image: vitsly / Hokusai

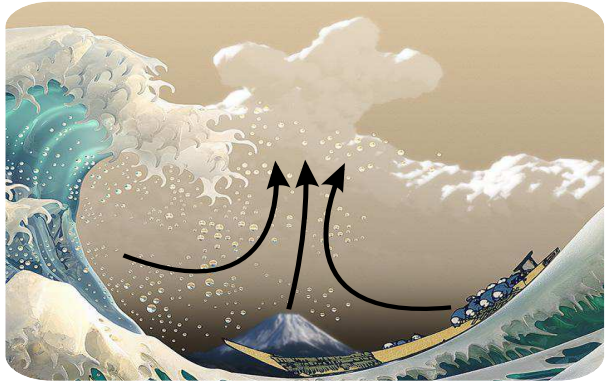
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- ▶ aerosol particles of natural and anthropogenic origin act as CCN/INP



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two-way interactions:

- aerosol characteristics influence cloud microstructure
- cloud processes influence aerosol size and composition

background image: vitsly / Hokusai

Aerosol-cloud interactions: μ -physics models

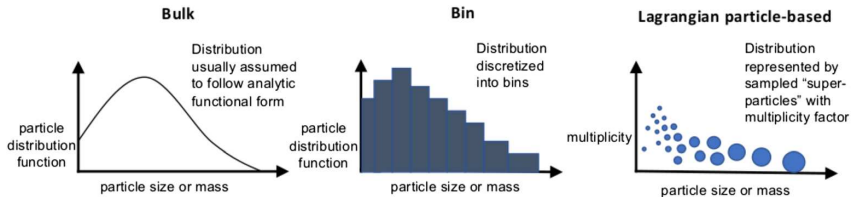


Figure 3. Representation of cloud and precipitation particle distributions in the three main types of microphysics
(Morrison et al. 2020, JAMES)

Aerosol-cloud interactions: μ -physics models

well suited for two-way aerosol-cloud interactions!

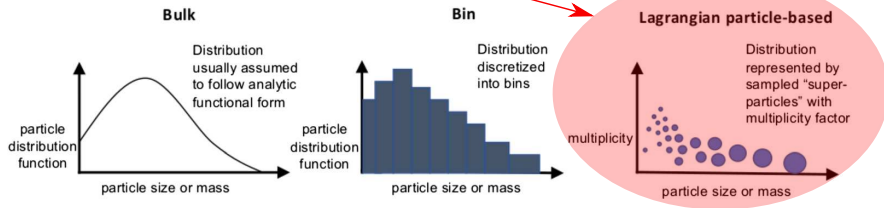


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DOI: [10.1175/BAMS-D-25-0026.1](https://doi.org/10.1175/BAMS-D-25-0026.1)

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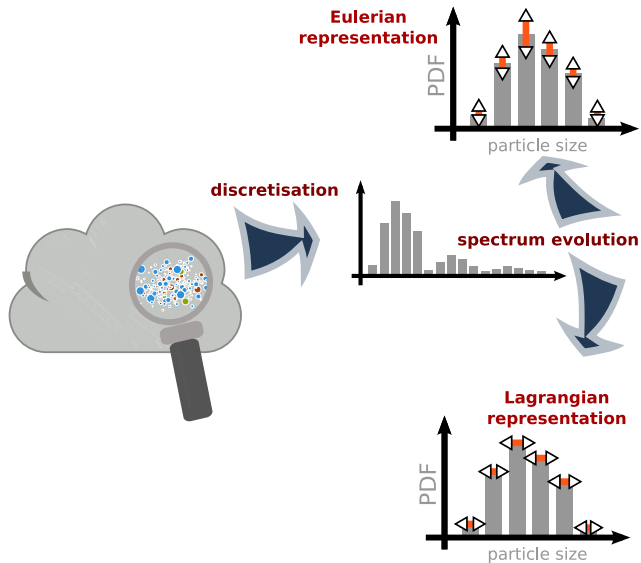
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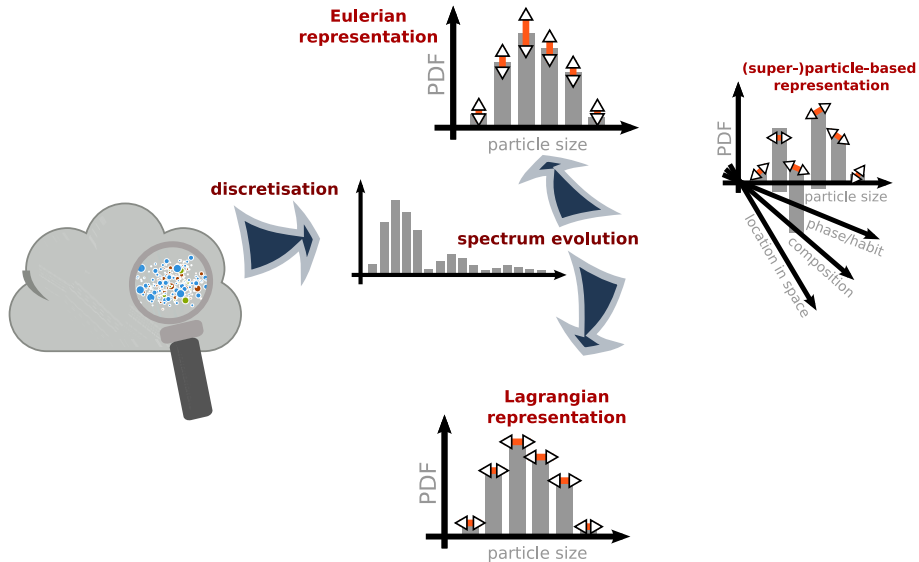
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- ▶ *... adding more predicted attributes, such as aerosol chemical constituents or particle density, with little computational cost ...*
- ▶ *suited for simulating cloud–aerosol interactions by tracking the aerosol budget explicitly ...”*

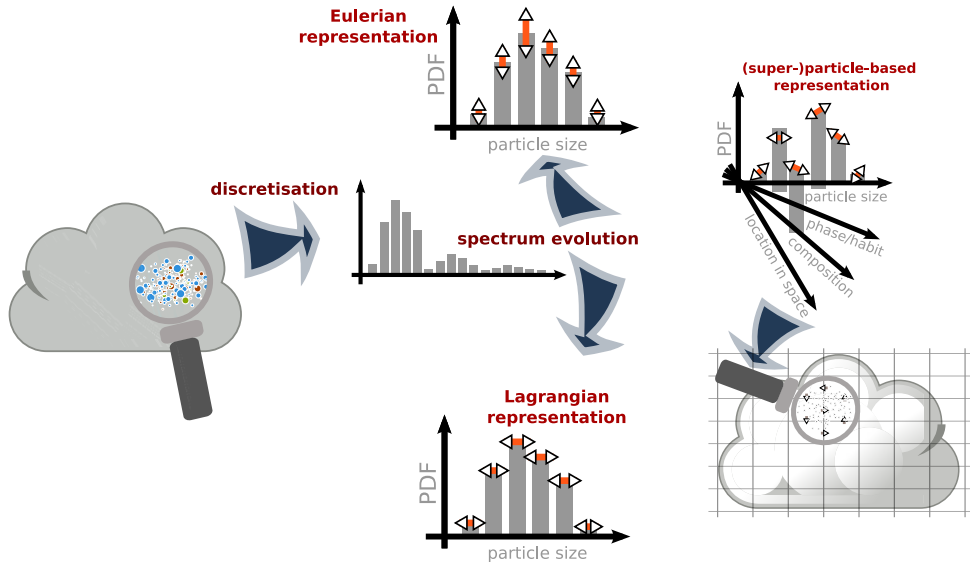
modelling cloud μ -physics: Eulerian vs. Lagrangian approaches



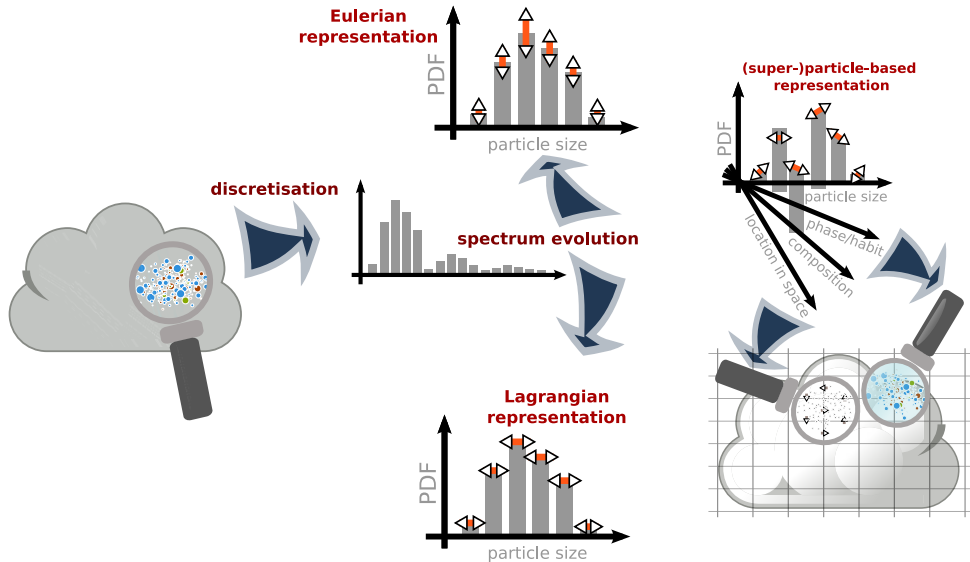
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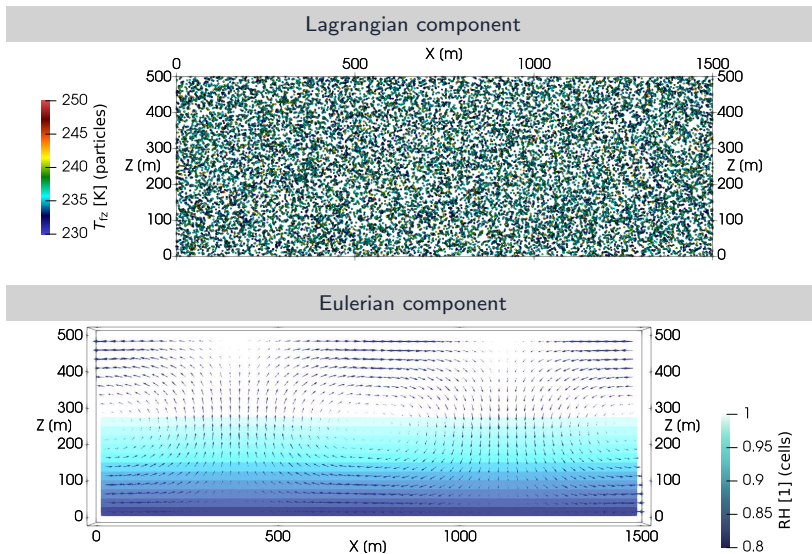
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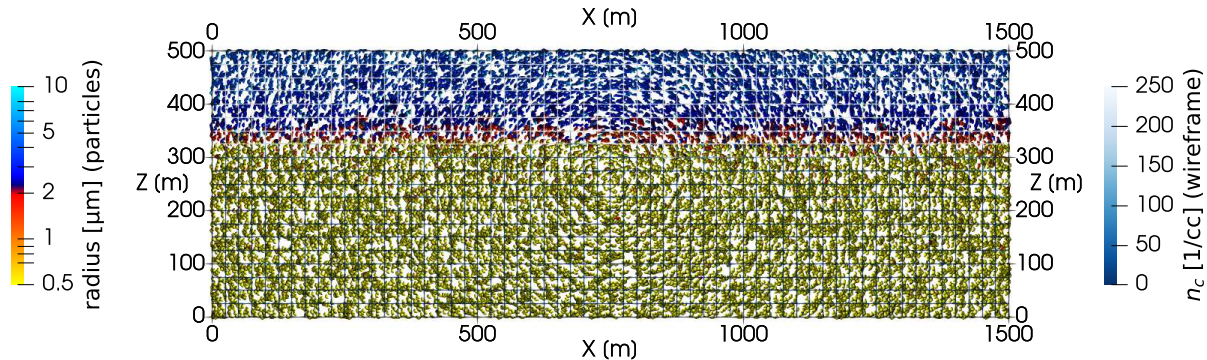


Particle-based μ -physics + prescribed-flow: model state



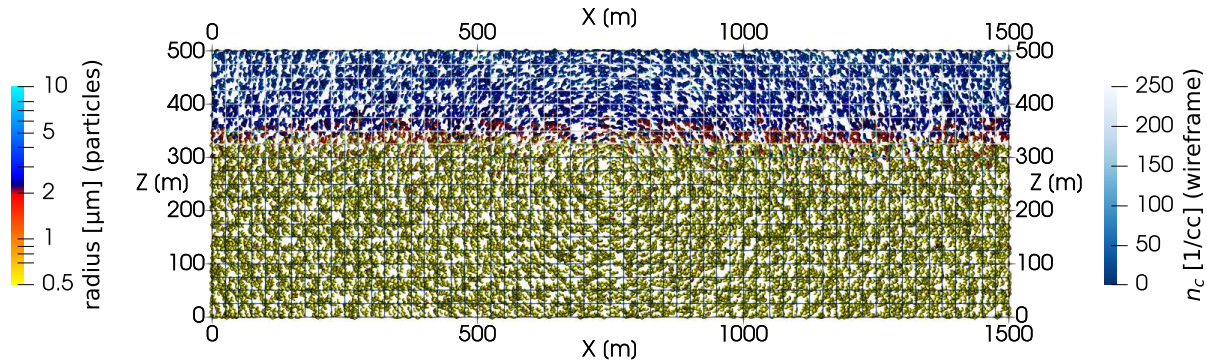
Particle-based μ -physics + prescribed-flow: spin-up

Time: 30 s (spin-up till 600.0 s)



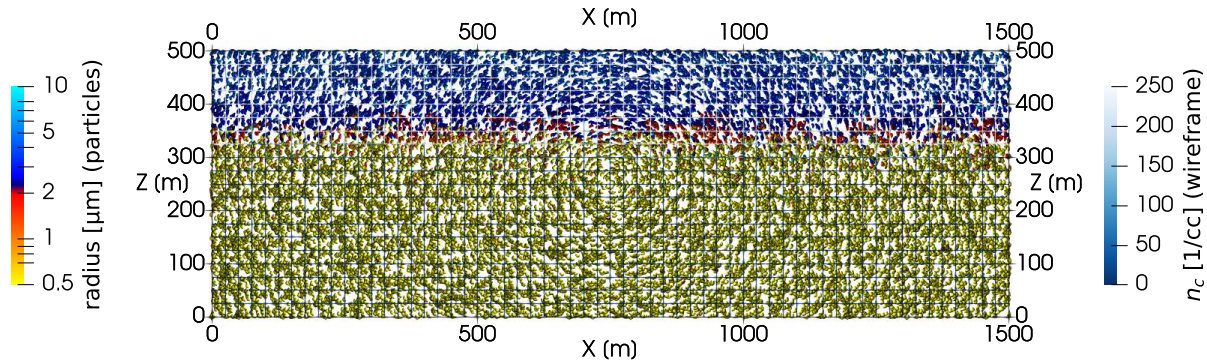
Particle-based μ -physics + prescribed-flow: spin-up

Time: 60 s (spin-up till 600.0 s)



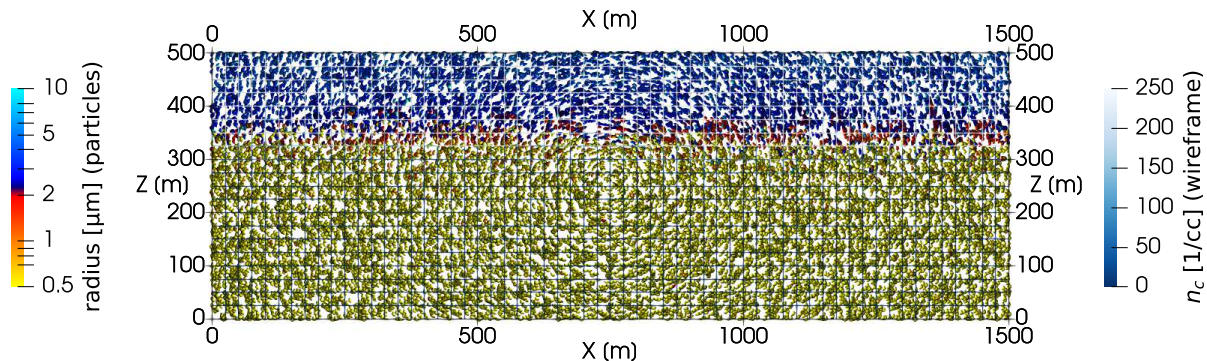
Particle-based μ -physics + prescribed-flow: spin-up

Time: 90 s (spin-up till 600.0 s)



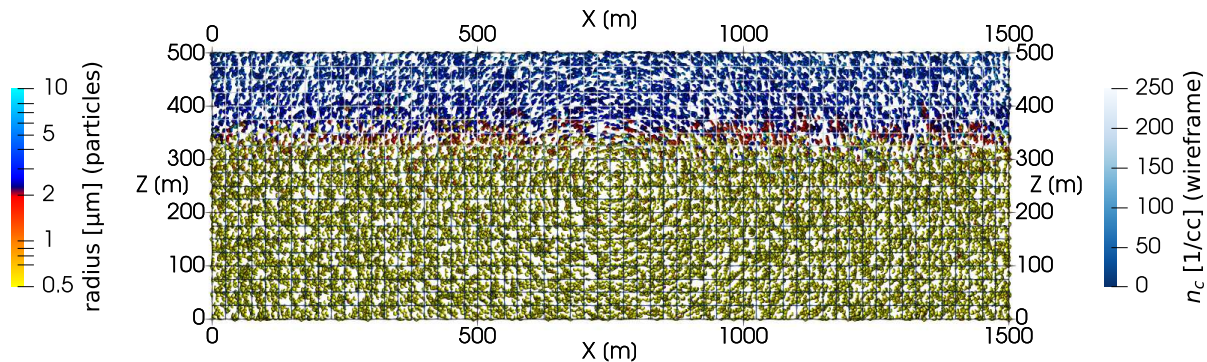
Particle-based μ -physics + prescribed-flow: spin-up

Time: 120 s (spin-up till 600.0 s)



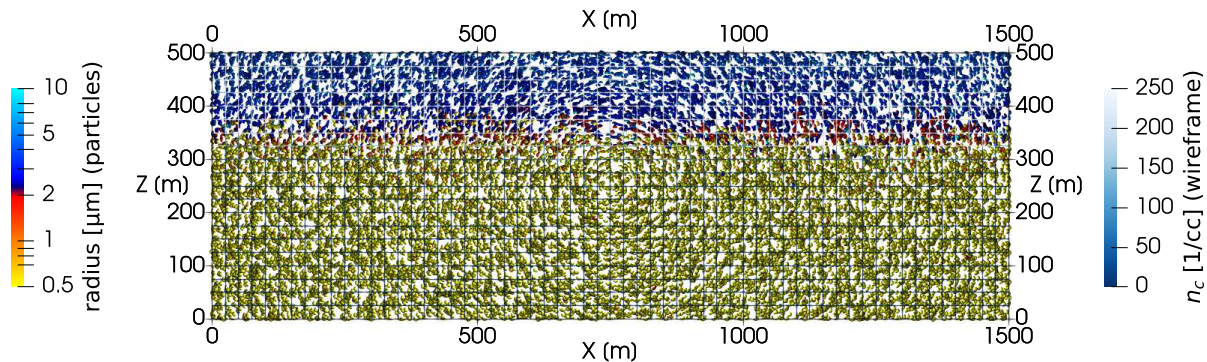
Particle-based μ -physics + prescribed-flow: spin-up

Time: 150 s (spin-up till 600.0 s)



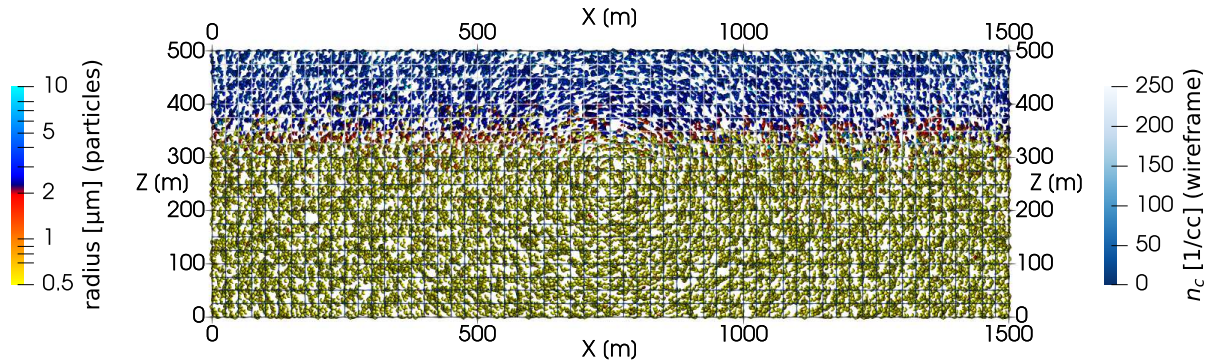
Particle-based μ -physics + prescribed-flow: spin-up

Time: 180 s (spin-up till 600.0 s)



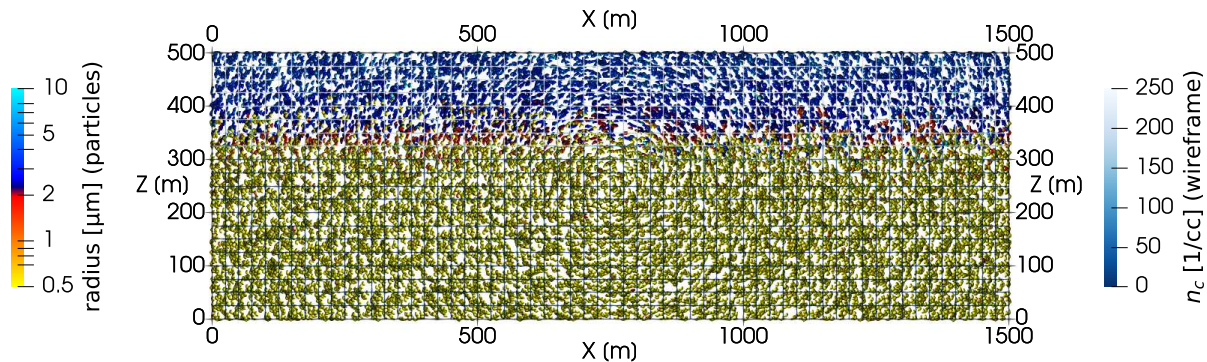
Particle-based μ -physics + prescribed-flow: spin-up

Time: 210 s (spin-up till 600.0 s)



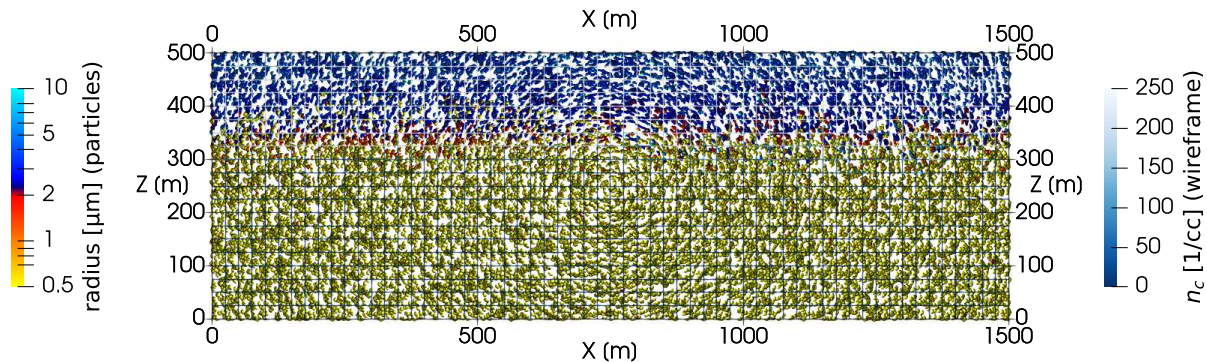
Particle-based μ -physics + prescribed-flow: spin-up

Time: 240 s (spin-up till 600.0 s)



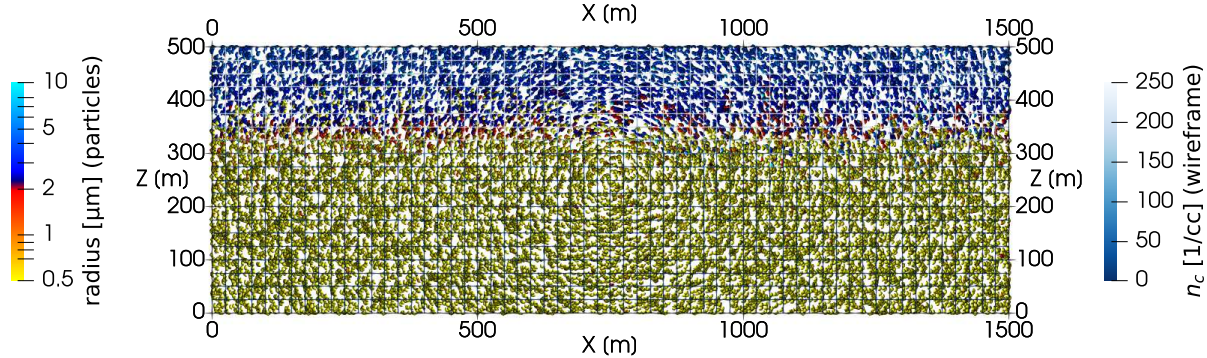
Particle-based μ -physics + prescribed-flow: spin-up

Time: 270 s (spin-up till 600.0 s)



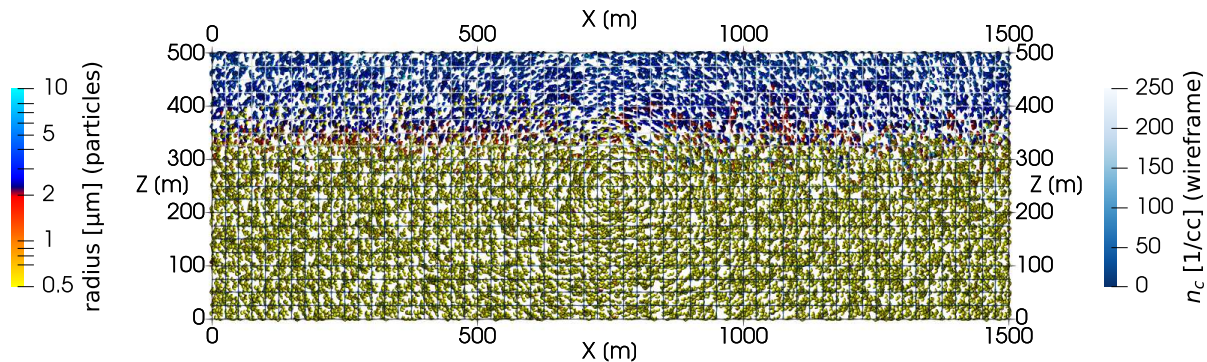
Particle-based μ -physics + prescribed-flow: spin-up

Time: 300 s (spin-up till 600.0 s)



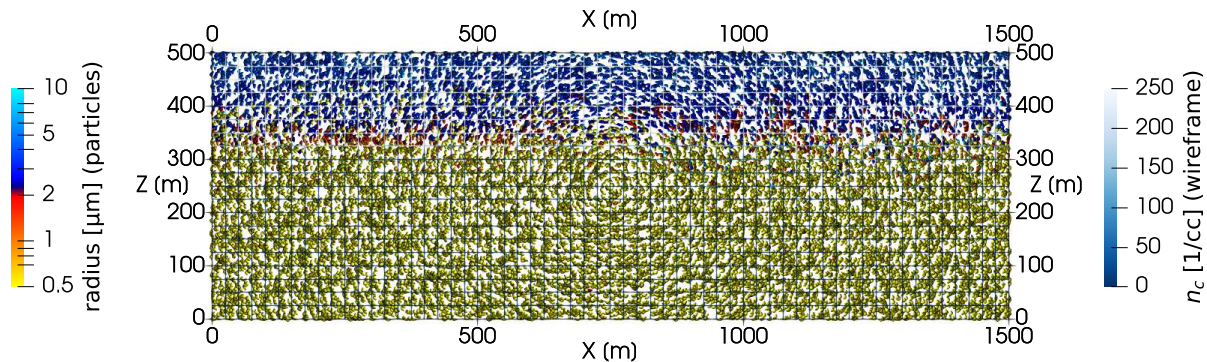
Particle-based μ -physics + prescribed-flow: spin-up

Time: 330 s (spin-up till 600.0 s)



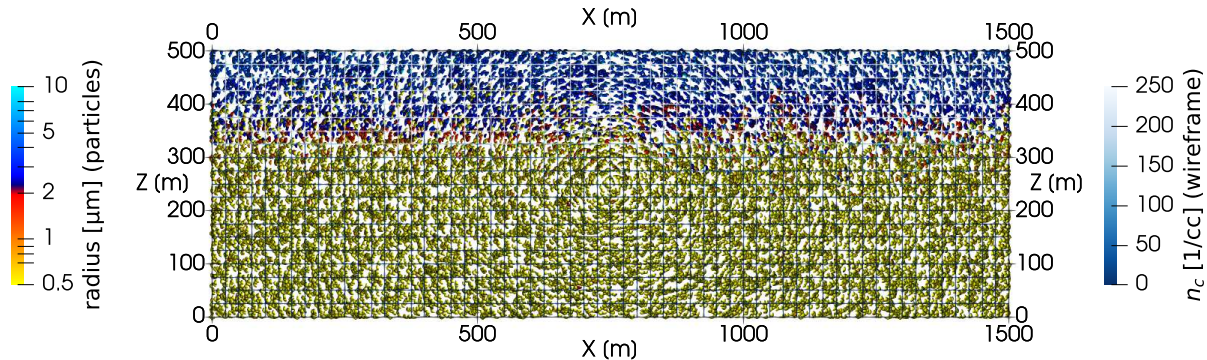
Particle-based μ -physics + prescribed-flow: spin-up

Time: 360 s (spin-up till 600.0 s)



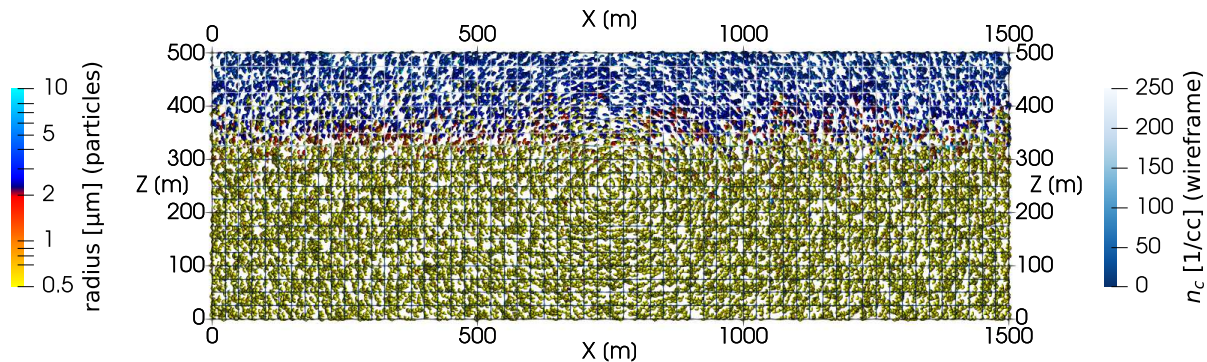
Particle-based μ -physics + prescribed-flow: spin-up

Time: 390 s (spin-up till 600.0 s)



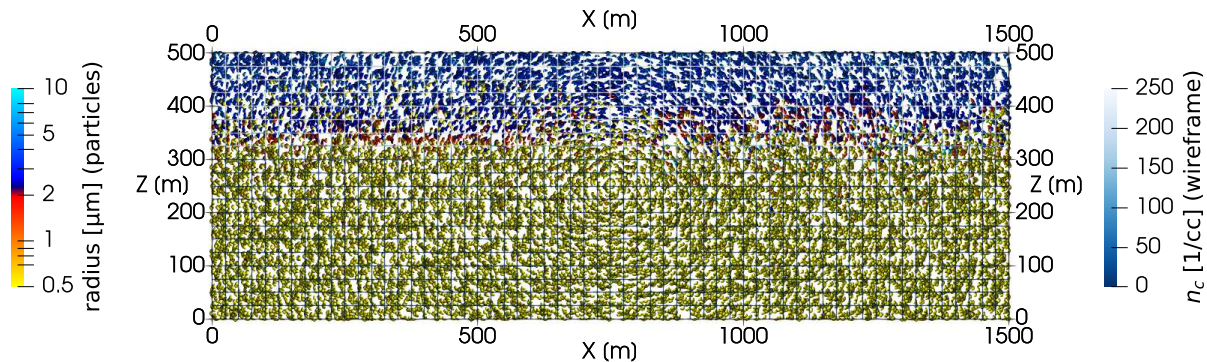
Particle-based μ -physics + prescribed-flow: spin-up

Time: 420 s (spin-up till 600.0 s)



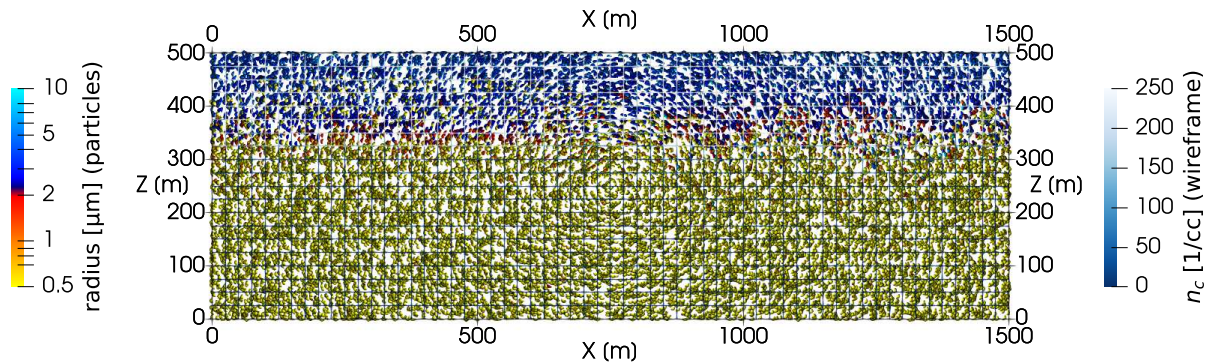
Particle-based μ -physics + prescribed-flow: spin-up

Time: 450 s (spin-up till 600.0 s)



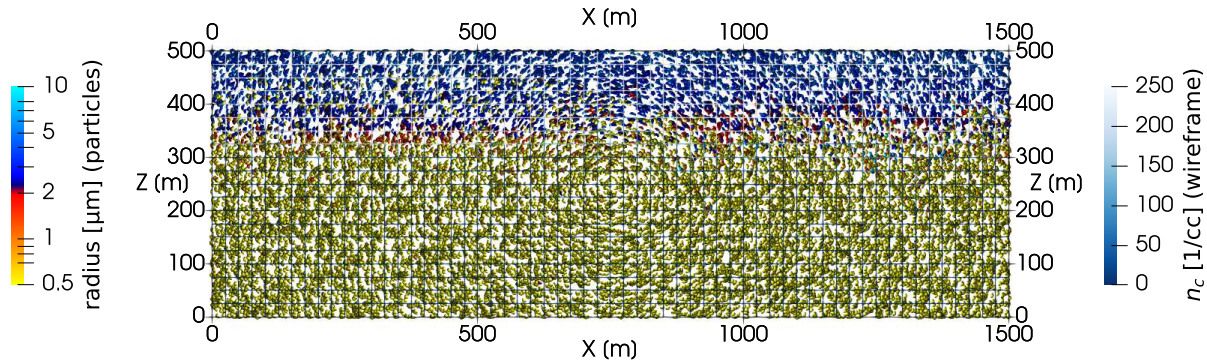
Particle-based μ -physics + prescribed-flow: spin-up

Time: 480 s (spin-up till 600.0 s)



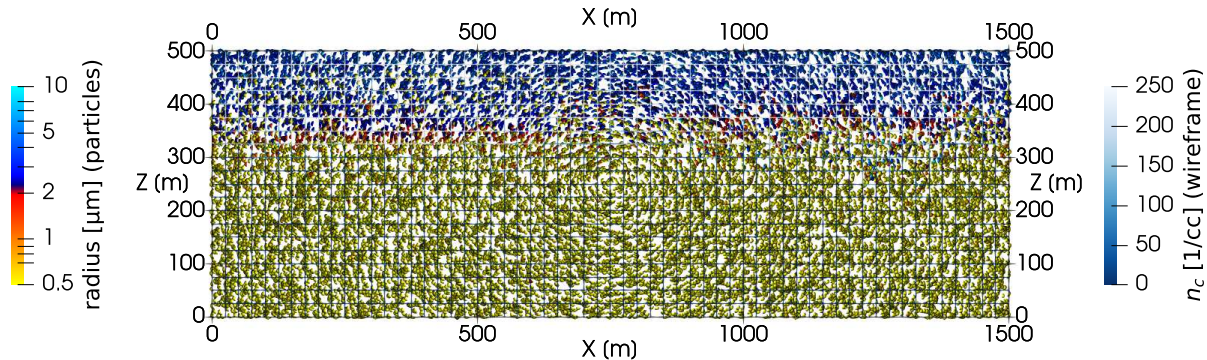
Particle-based μ -physics + prescribed-flow: spin-up

Time: 510 s (spin-up till 600.0 s)



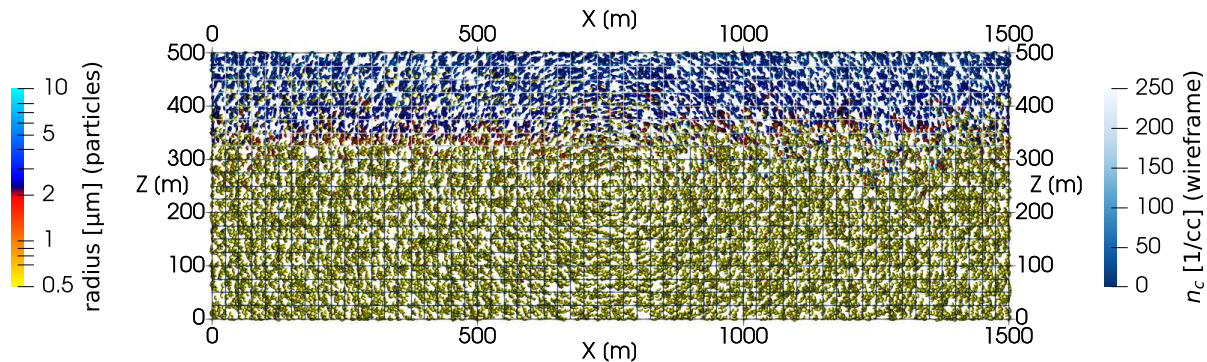
Particle-based μ -physics + prescribed-flow: spin-up

Time: 540 s (spin-up till 600.0 s)



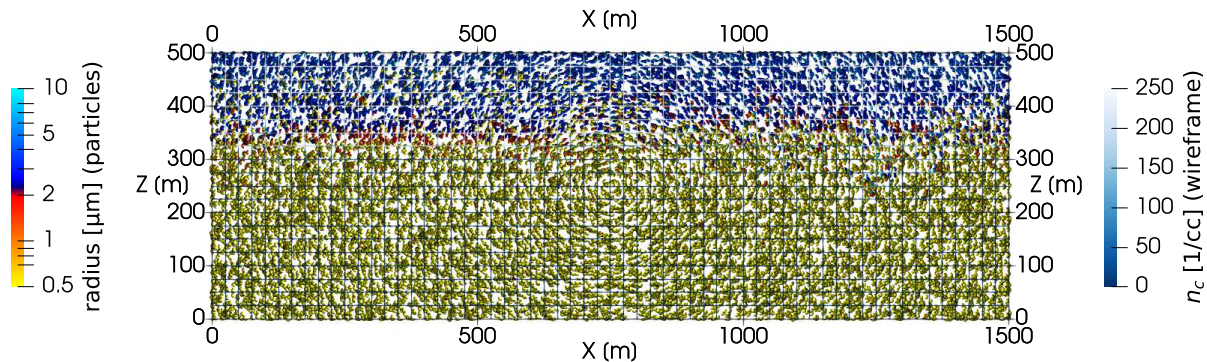
Particle-based μ -physics + prescribed-flow: spin-up

Time: 570 s (spin-up till 600.0 s)



Particle-based μ -physics + prescribed-flow: spin-up

Time: 600 s (spin-up till 600.0 s)



Shima, Sato, Hashimoto & Misumi 2020 (GMD):

Predicting the morphology of ice particles in deep convection using the super-droplet method

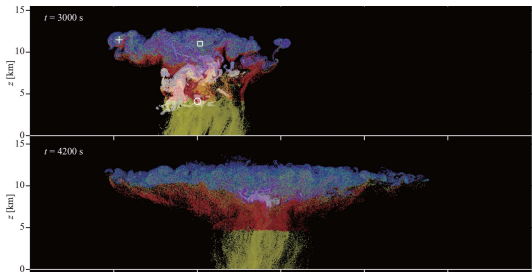


Figure 1. Typical realization of CTRL cloud spatial structures at $t = 2040, 2460, 3000, 4200$, and 5400 s. The mixing ratio of cloud water, rainwater, cloud ice, graupel, and snow aggregates are plotted in fading white, yellow, blue, red, and green, respectively. The symbols indicate examples of unrealistic predicted ice particles (Sects. 7.3 and 9.1). See also Movie 1 in the video supplement.

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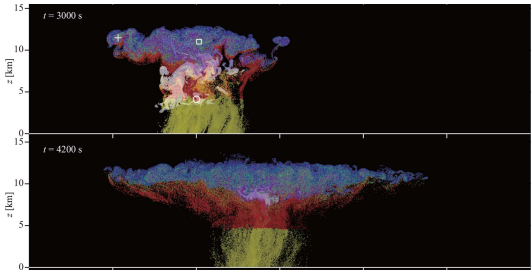


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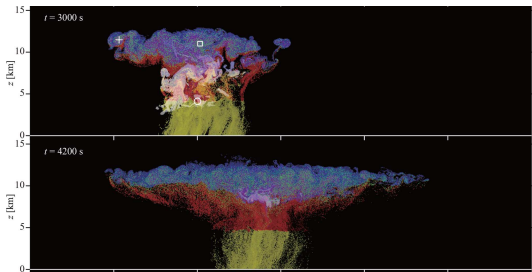


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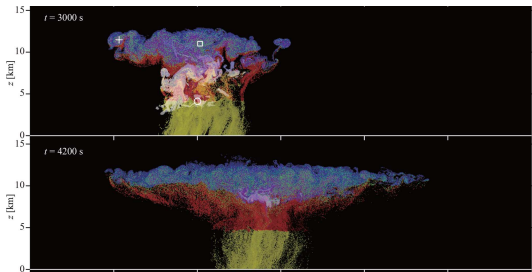


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- ▶ particle-resolved processes:
 - advection and sedimentation
 - homogeneous and immersion freezing (singular)
 - melting
 - condensation and evaporation (incl. CCN [de]activation)
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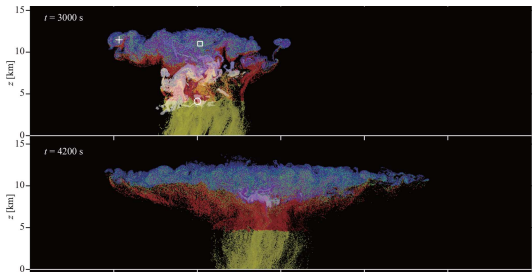


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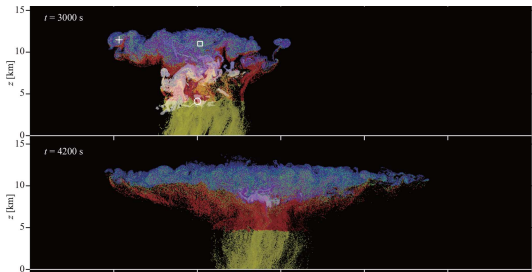


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- ▶ 2D Cb test case

singular $\rightsquigarrow$ #INP vs. temperature

Vali 2014 (ACP)

“emphasis in experimental studies is placed on measurements of the temperature dependence and/or of the time dependence of nucleation.

Interpretations of the experimental results face considerable difficulties in identifying the relative importance of the two factors

... dichotomy – the stochastic and singular models”

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common underlying Poissonian model

Poissonian model of freezing & Ice Nucleation Active Sites (INAS)

theory (in modern notation)

(Bigg '53, Langham & Mason '58, Carte '59, Marshall '61)

Poisson counting process with rate r :

$$P^*(k \text{ events in time } t) = \frac{(rt)^k \exp(-rt)}{k!}$$

$$P(\text{one or more events in time } t) = 1 - P^*(k = 0, t)$$

$$\ln(1 - P) = -rt$$

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introducing $J_{\text{het}}(T)$, $T(t)$ and INP surface A :

$$\ln(1 - P(A, t)) = -A \underbrace{\int_0^t J_{\text{het}}(T(t')) dt'}_{\text{ }}$$

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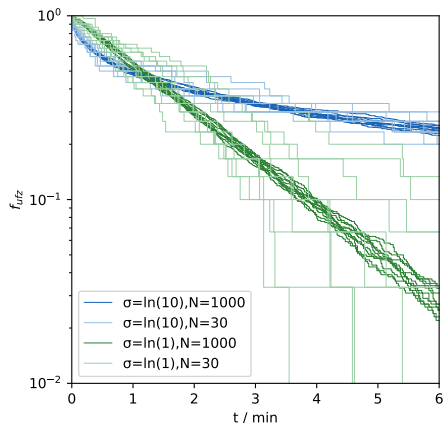
$$P(\text{one or more events in time } t) = 1 - P^*(k = 0, t)$$

$$\ln(1 - P) = -rt$$

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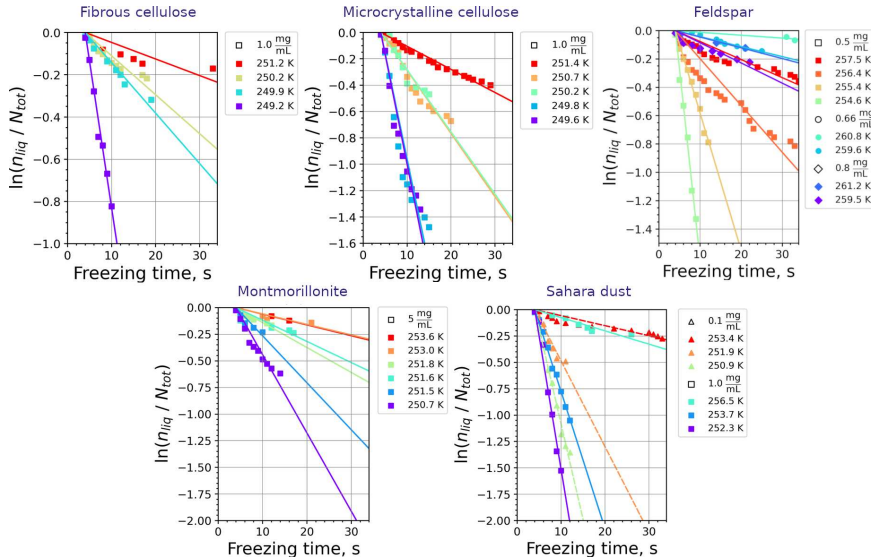
$$\ln(1 - P(A, t)) = -A \underbrace{\int_0^t J_{\text{het}}(T(t')) dt'}_{\text{ }}$$

Monte Carlo: const J_{het} , lognormal A



(as in Alpert & Knopf 2016, Fig. 1a)

Szakáll et al. 2021, ACP 21: isothermal experiments



Poissonian model of freezing & Ice Nucleation Active Sites (INAS)

theory (in modern notation)

(Bigg '53, Langham & Mason '58, Carte '59, Marshall '61)

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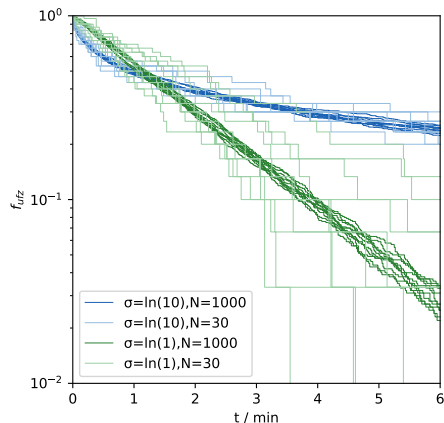
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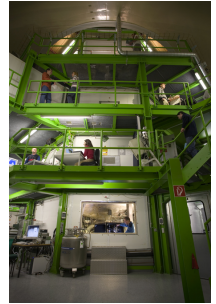
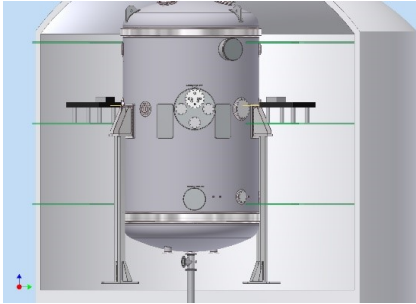
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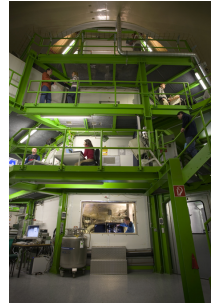
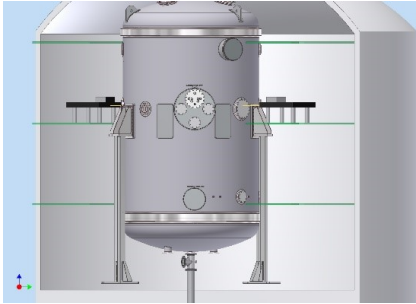
experimental $n_s(T)$ fits: e.g., Niemand et al. 2012

AIDA @ KIT



(<https://www.imk-aaf.kit.edu/>, photo: KIT/Ottmar Möhler)

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(<https://www.imk-aaf.kit.edu/>, photo: KIT/Ottmar Möhler)

AIDA cooling rate: ca. 0.5 K/min

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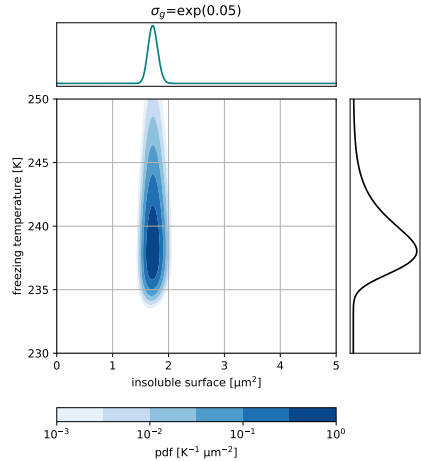
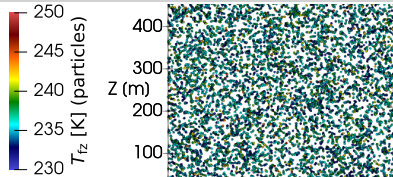
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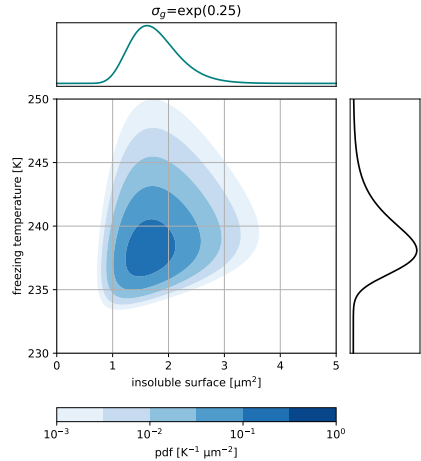
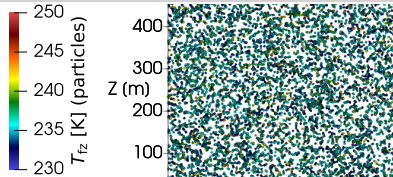
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singular n_s -based (INAS)



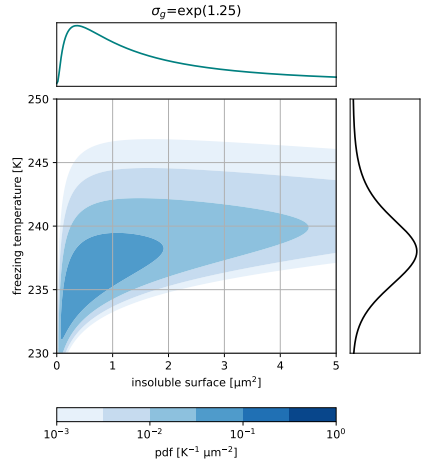
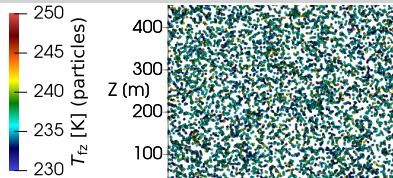
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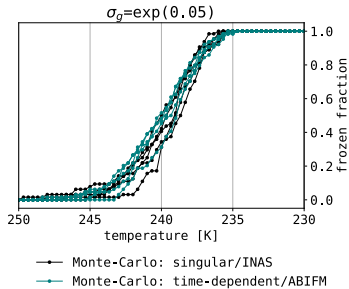
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time-dependent: A as **attribute** (randomly sampled from the same lognormal)
Monte-Carlo freezing trigger using $P(A \cdot J_{het}(T_{ambient}(t)))$

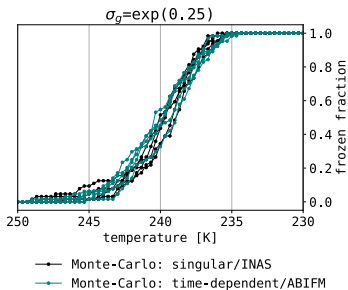
AIDA cooling rate: 0.5 K/min



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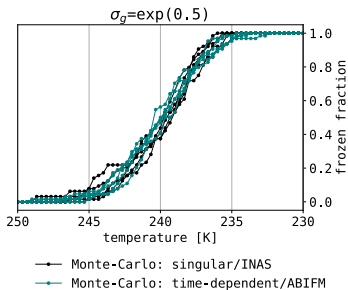
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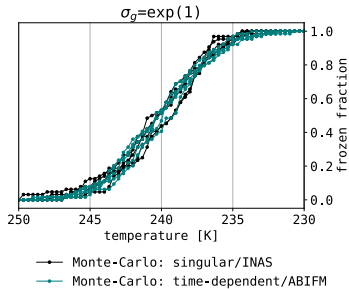
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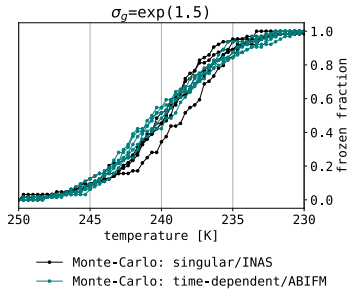
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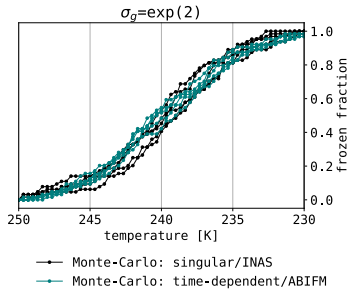
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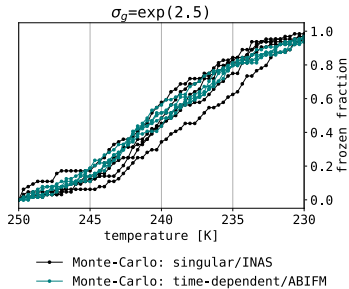
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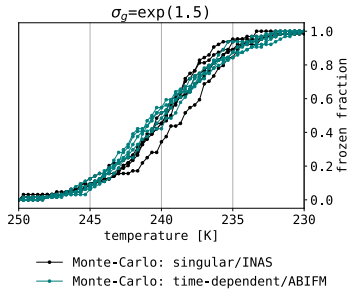
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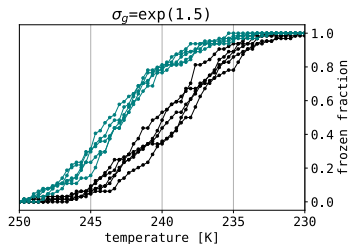
AIDA cooling rate: 0.5 K/min



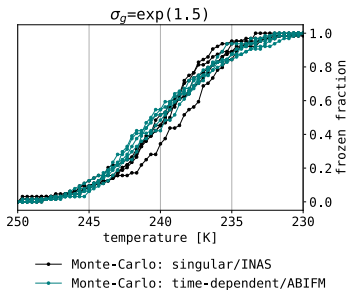
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cooling rate: 0.1 K/min



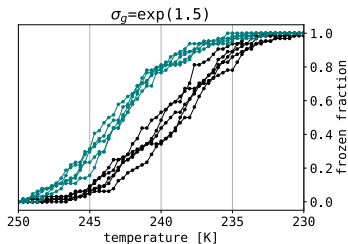
AIDA cooling rate: 0.5 K/min



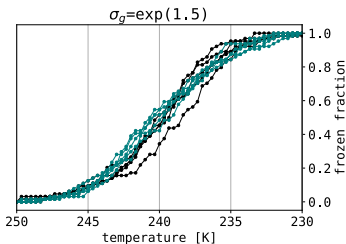
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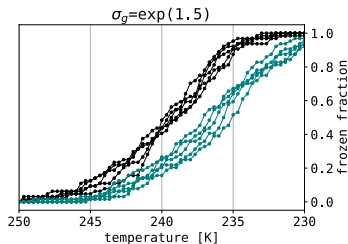


AIDA cooling rate: 0.5 K/min



—●— Monte-Carlo: singular/INAS
—●— Monte-Carlo: time-dependent/ABIFM

cooling rate: 2.5 K/min



Poissonian model of freezing & Ice Nucleation Active Sites (INAS)

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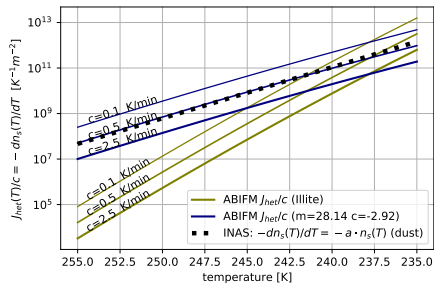
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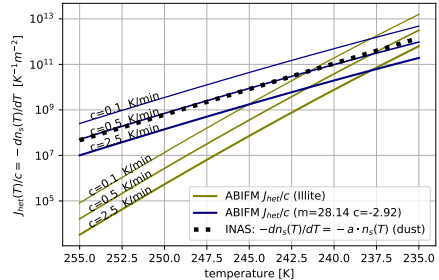
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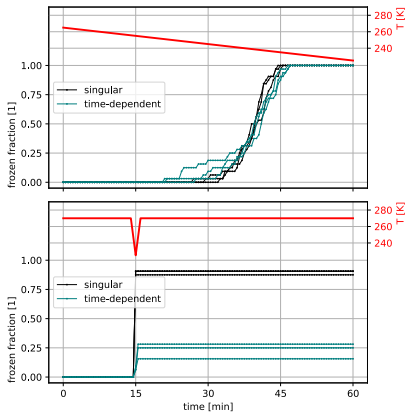
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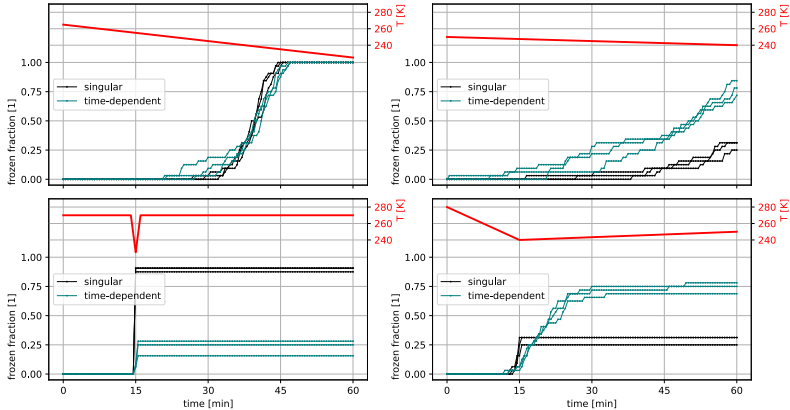
Is it a problem?



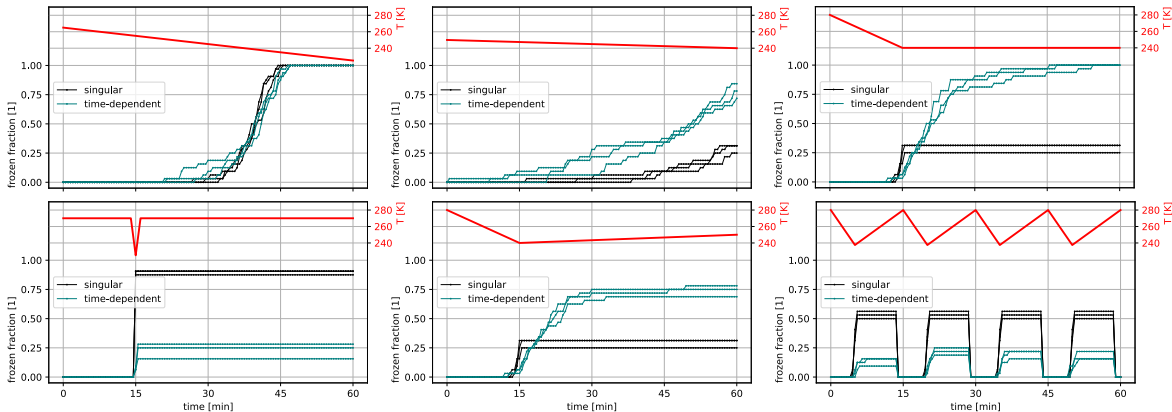
Testing different cooling-rate profiles in a box model



Testing different cooling-rate profiles in a box model

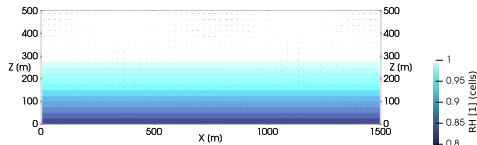


Testing different cooling-rate profiles in a box model



Testing three flow regimes and two immersion freezing representations

$w_{\max} \approx 1/3 \text{ m/s}$

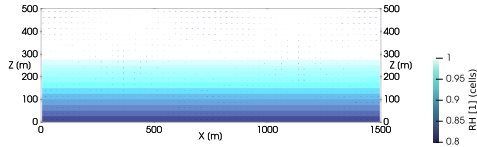


$w_{\max} \approx 1 \text{ m/s}$

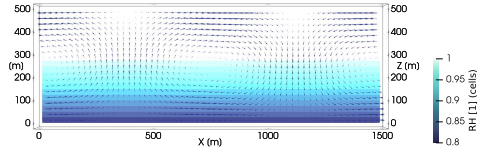
$w_{\max} \approx 3 \text{ m/s}$

Testing three flow regimes and two immersion freezing representations

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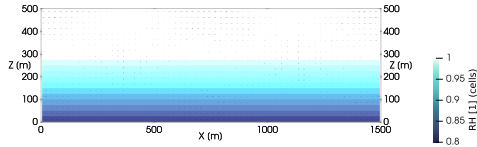
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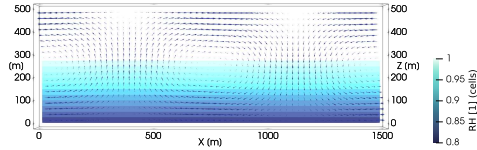
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Testing three flow regimes and two immersion freezing representations

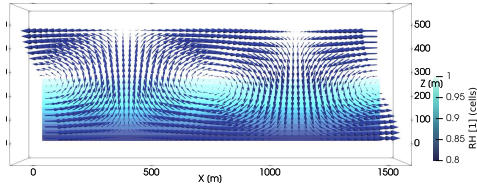
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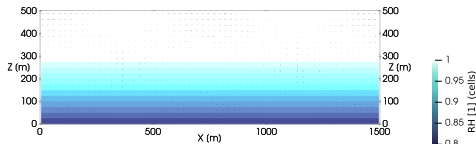


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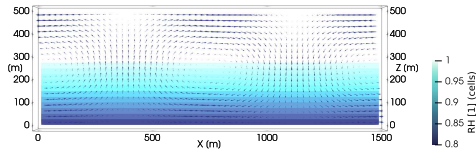


Testing three flow regimes and two immersion freezing representations

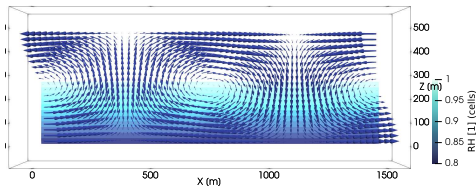
$w_{\max} \approx 1/3 \text{ m/s}$



$w_{\max} \approx 1 \text{ m/s}$

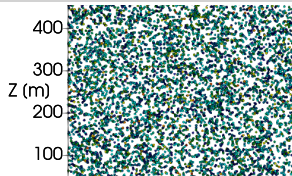


$w_{\max} \approx 3 \text{ m/s}$



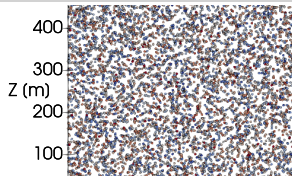
singular (INAS)

$T_{fz} \text{ [K (particles)]}$



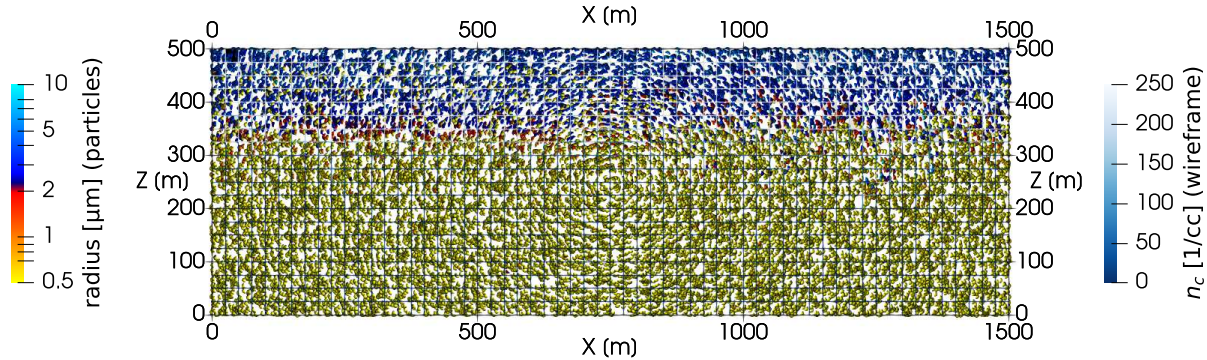
time-dependent (J_{het})

$A \text{ [}\mu\text{m}^2\text{]}$



Particle-based μ -physics + prescribed-flow: glaciation

Time: 630 s (spin-up till 600.0 s)



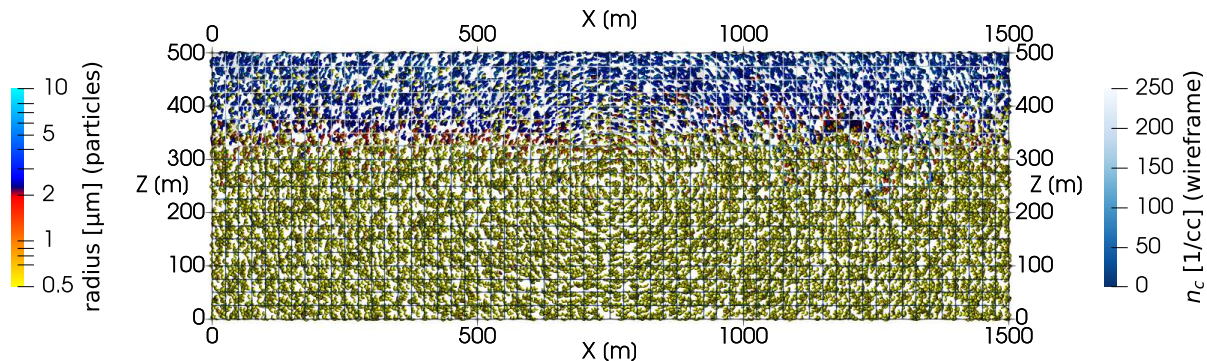
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 660 s (spin-up till 600.0 s)



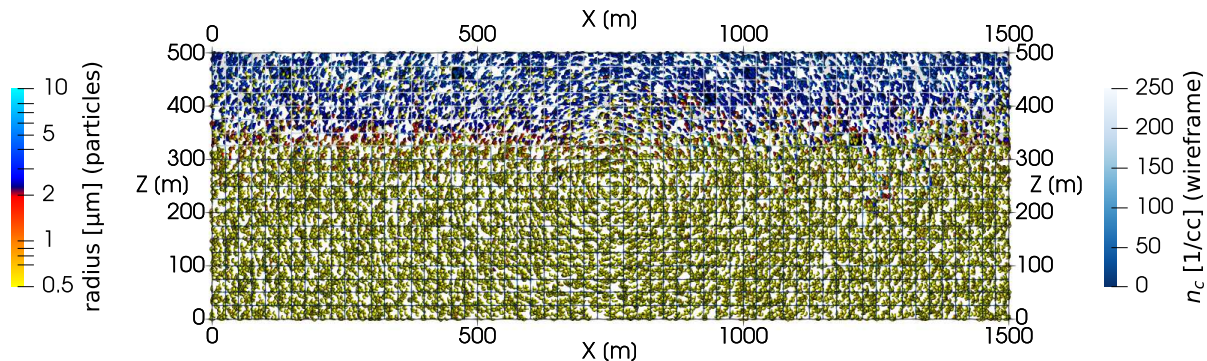
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 690 s (spin-up till 600.0 s)



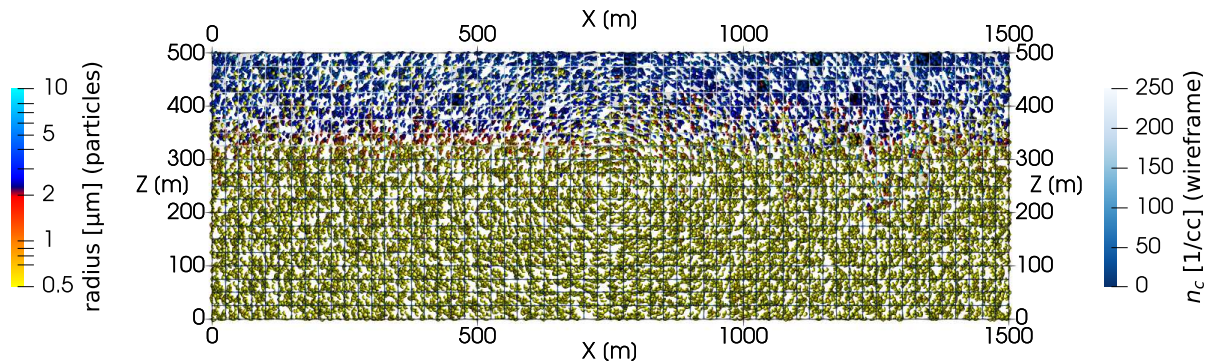
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 720 s (spin-up till 600.0 s)



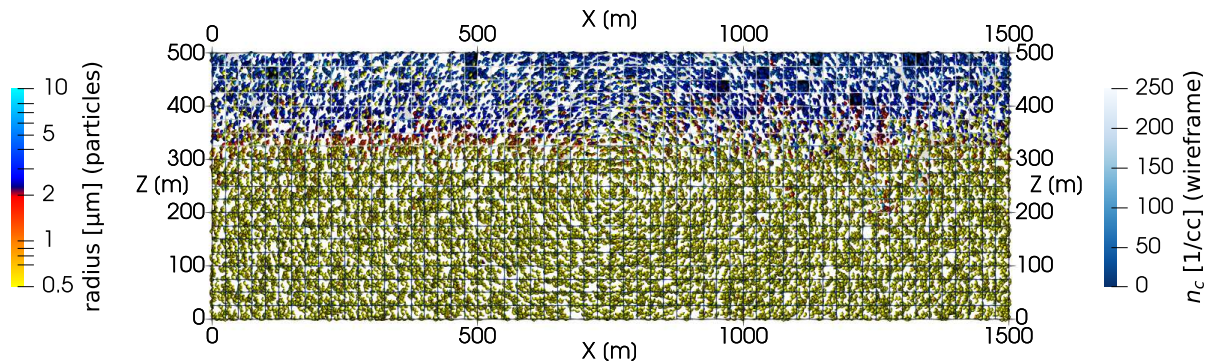
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 750 s (spin-up till 600.0 s)



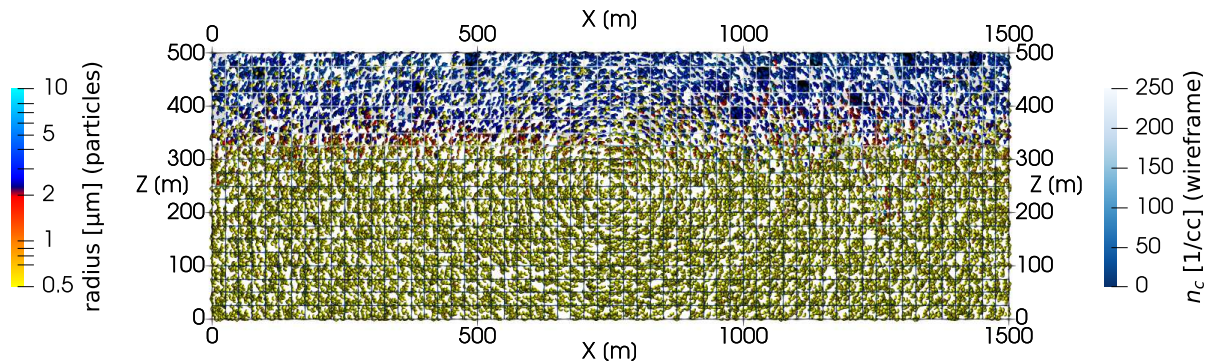
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 780 s (spin-up till 600.0 s)



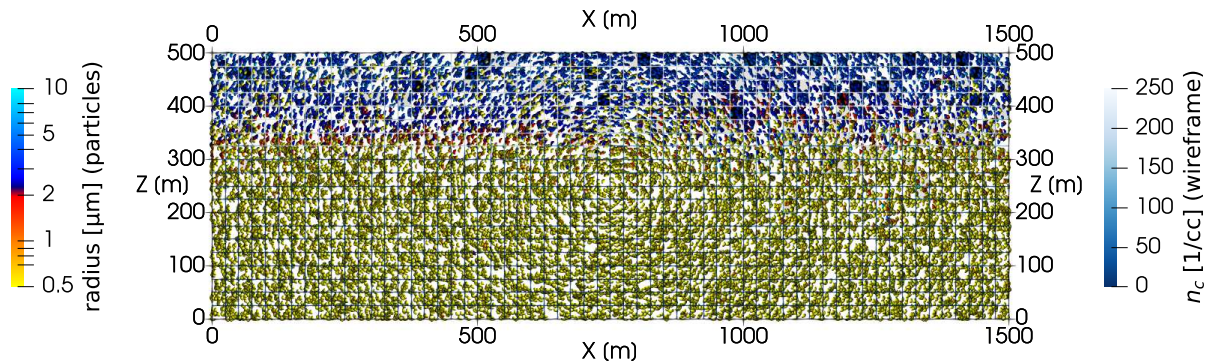
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 810 s (spin-up till 600.0 s)



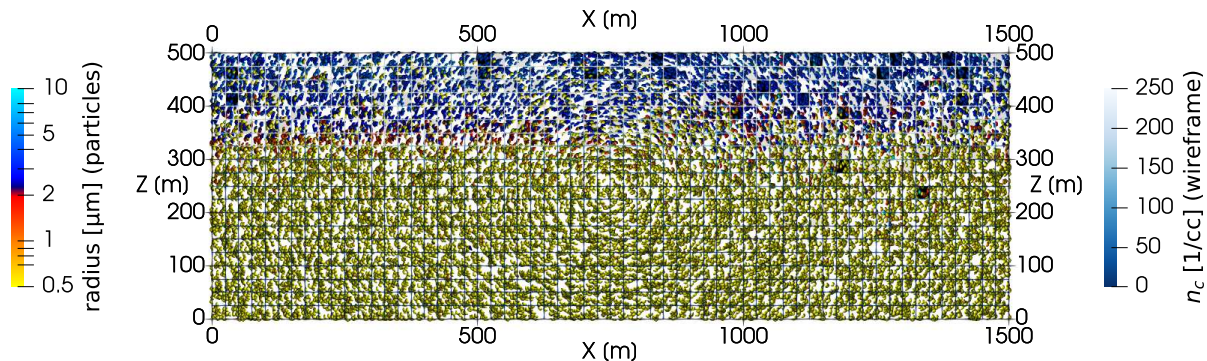
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 840 s (spin-up till 600.0 s)



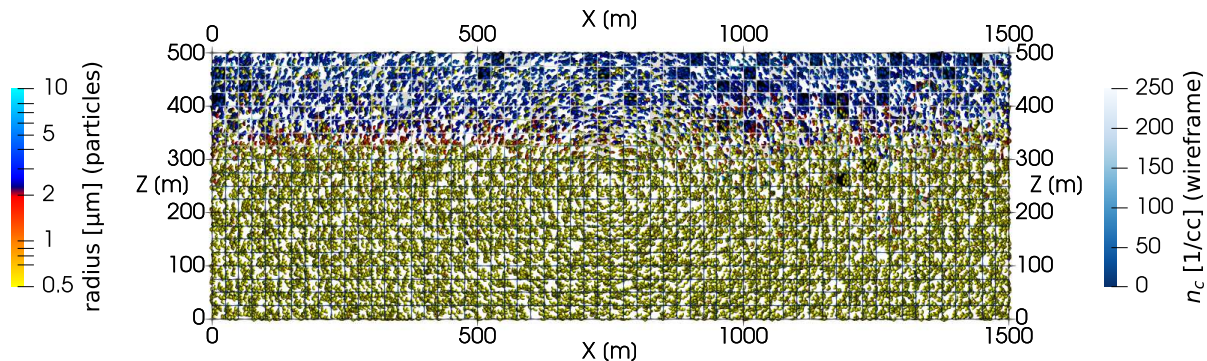
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 870 s (spin-up till 600.0 s)



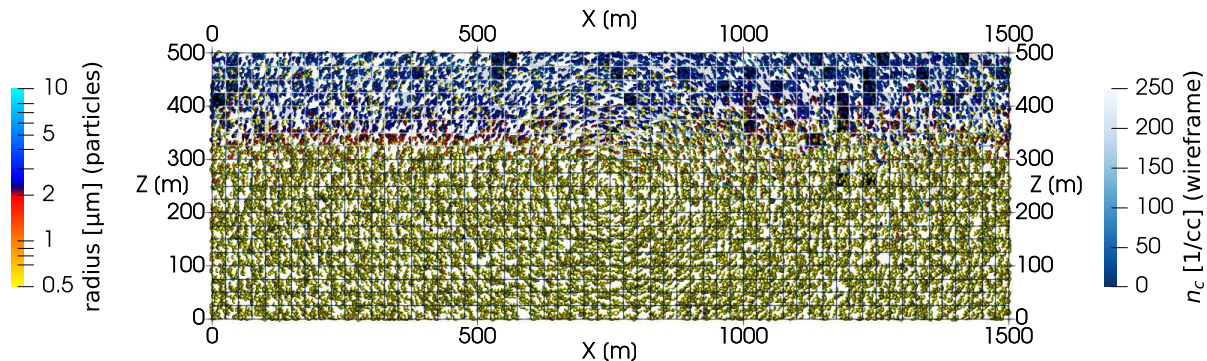
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 900 s (spin-up till 600.0 s)



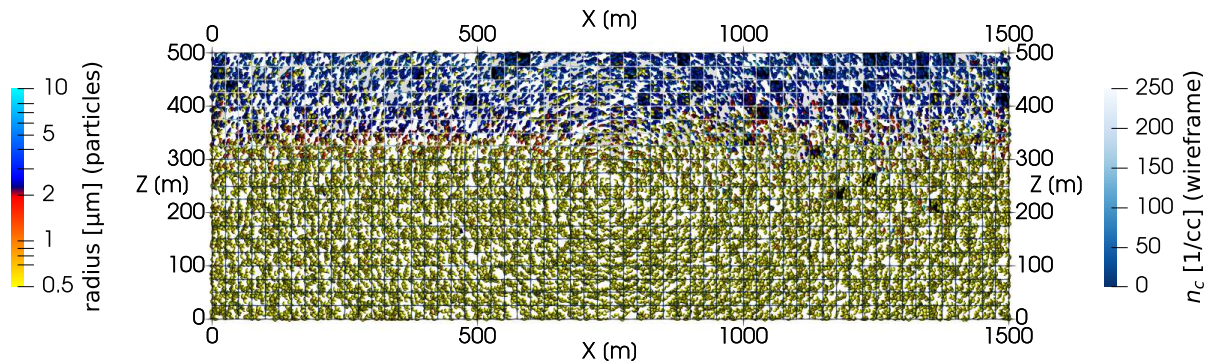
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 930 s (spin-up till 600.0 s)



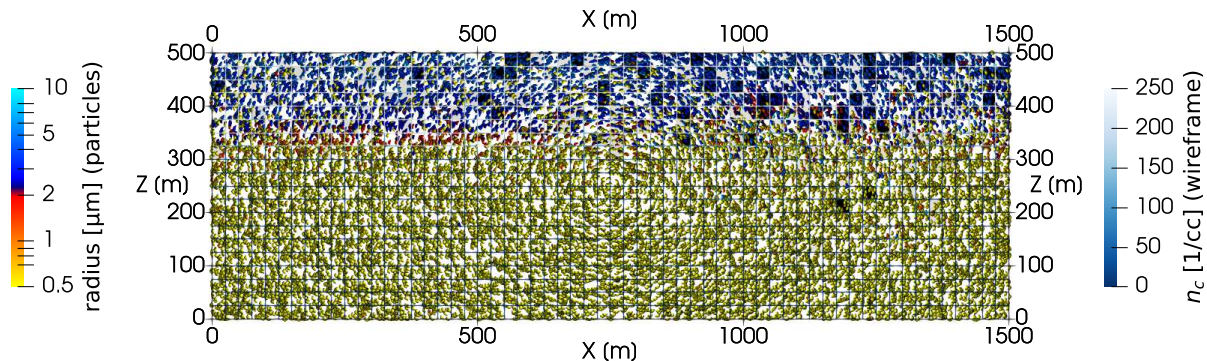
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 960 s (spin-up till 600.0 s)



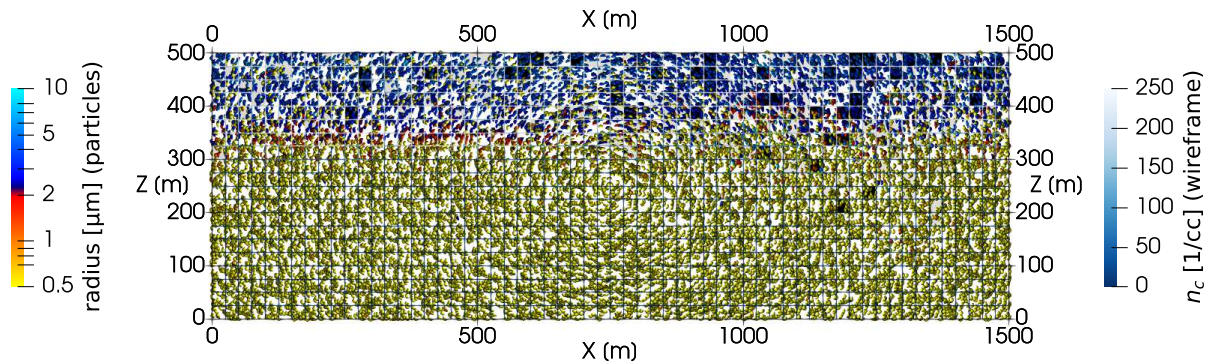
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 990 s (spin-up till 600.0 s)



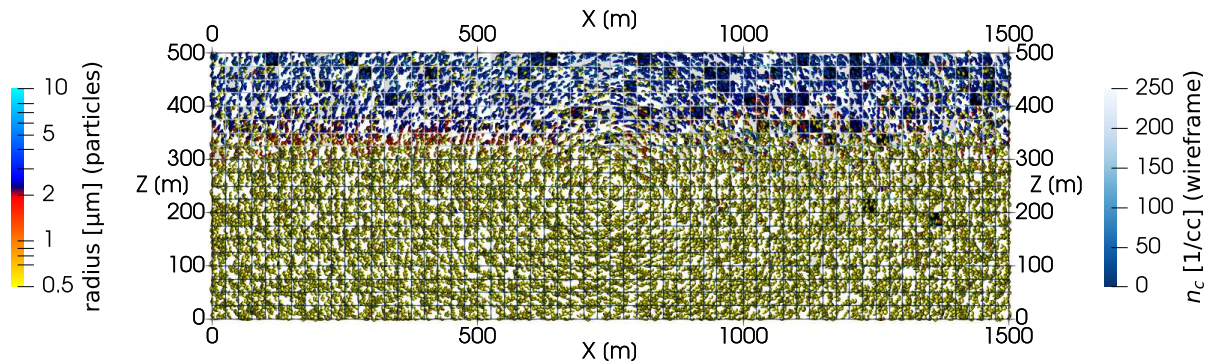
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1020 s (spin-up till 600.0 s)



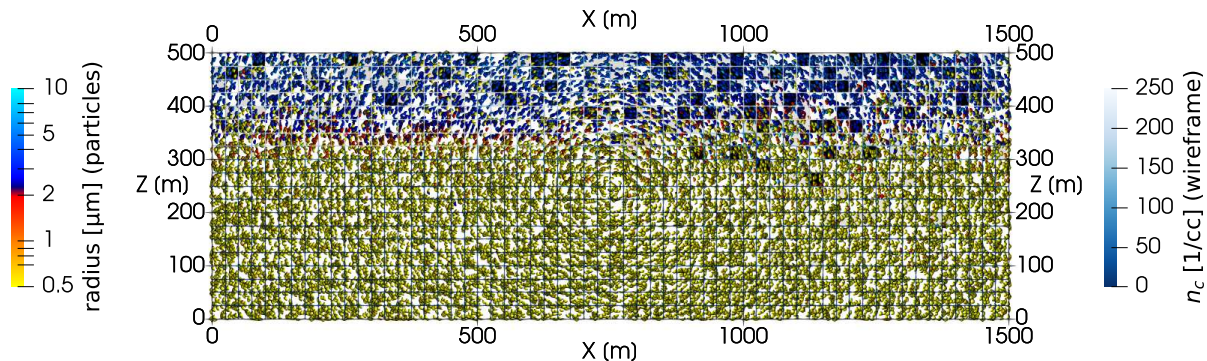
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1050 s (spin-up till 600.0 s)



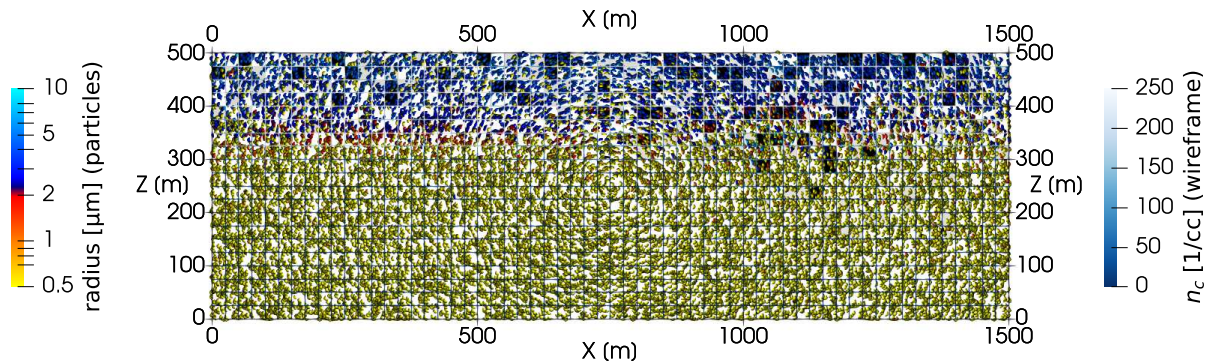
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1080 s (spin-up till 600.0 s)



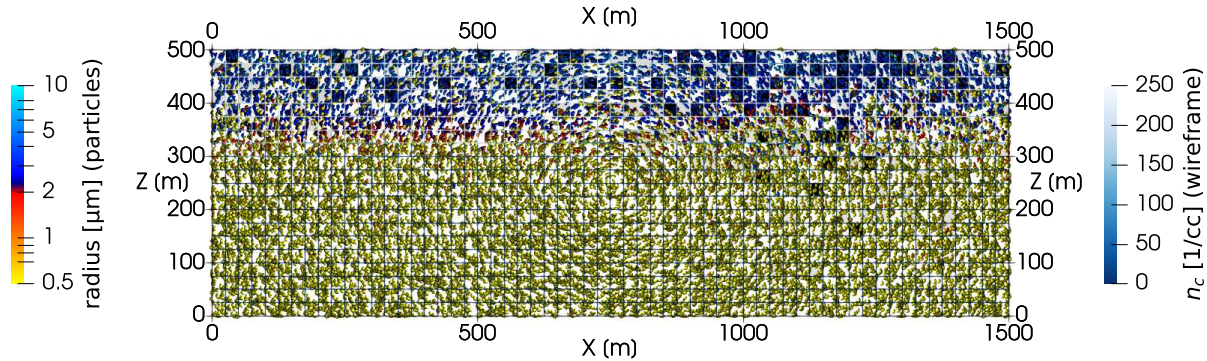
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1110 s (spin-up till 600.0 s)



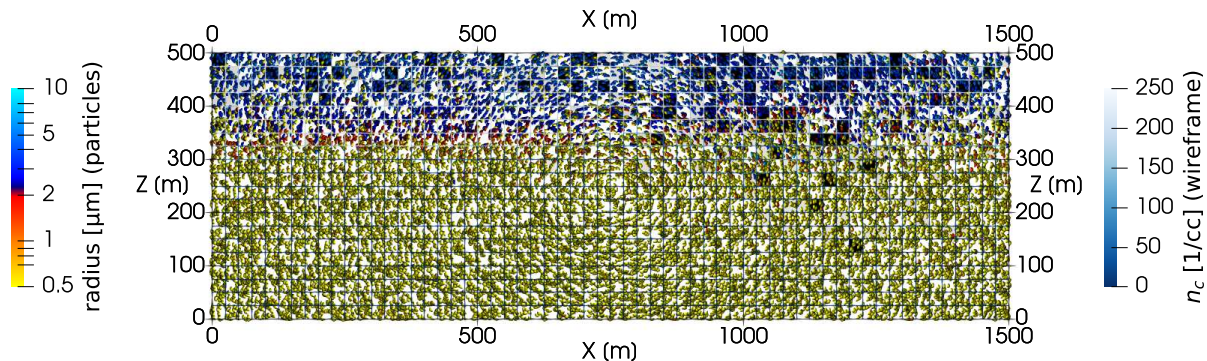
16+16 super-particles/cell for INP-rich + INP-free particles

$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1140 s (spin-up till 600.0 s)



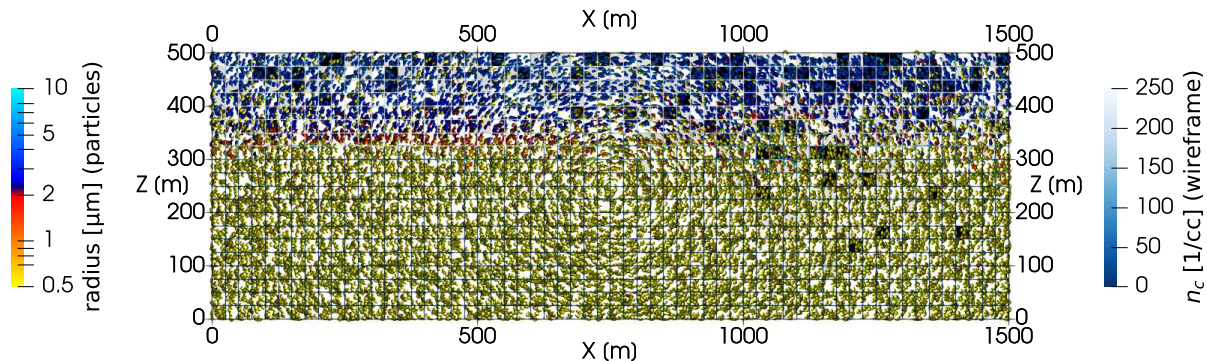
16+16 super-particles/cell for INP-rich + INP-free particles

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spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1170 s (spin-up till 600.0 s)



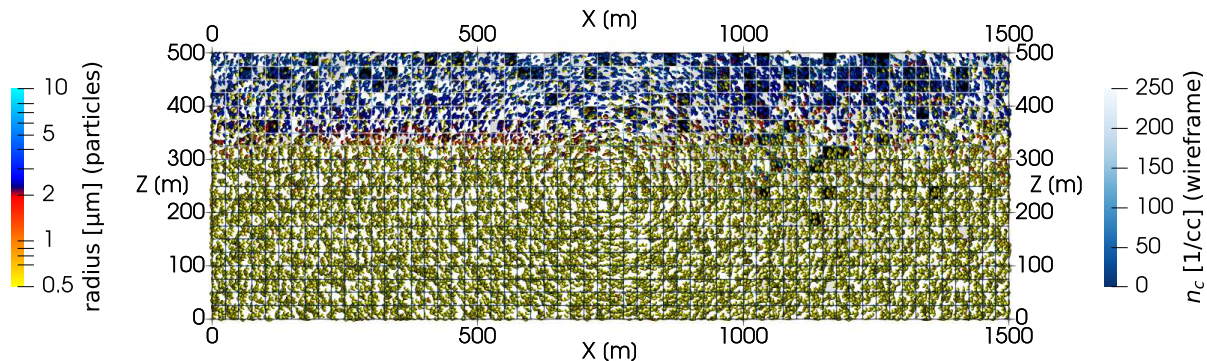
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$N_{\text{aer}} = 300/\text{cc}$ (two-mode lognormal) $N_{\text{INP}} = 150/L$ (lognormal, $D_g = 0.74 \mu\text{m}$, $\sigma_g = 2.55$)

spin-up = freezing off; subsequently frozen particles act as tracers

Particle-based μ -physics + prescribed-flow: glaciation

Time: 1200 s (spin-up till 600.0 s)

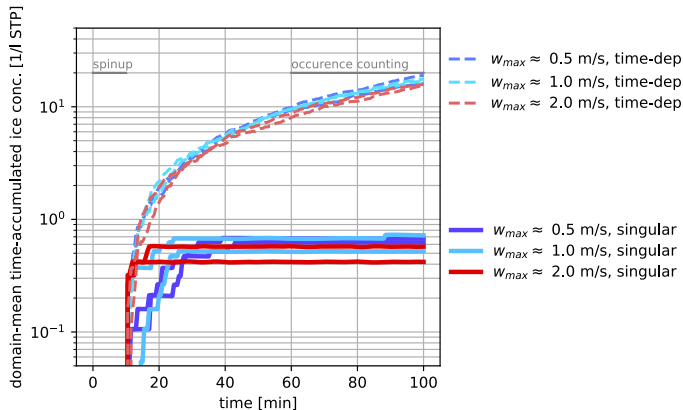


16+16 super-particles/cell for INP-rich + INP-free particles

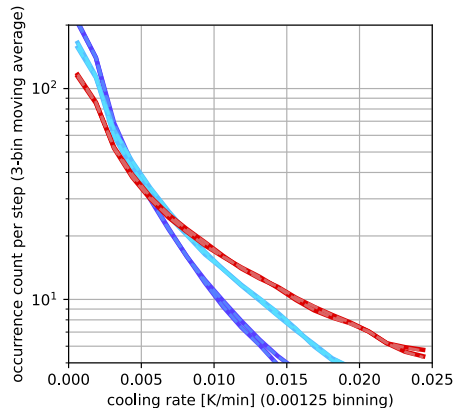
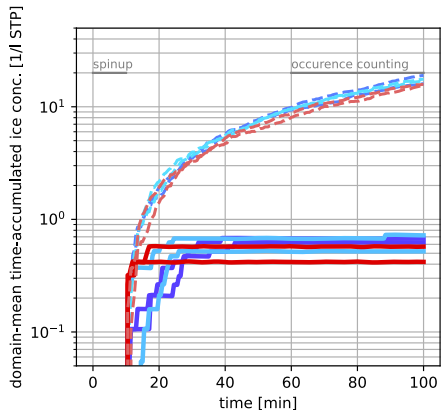
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spin-up = freezing off; subsequently frozen particles act as tracers

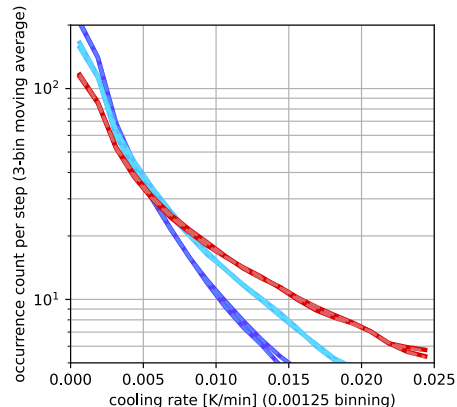
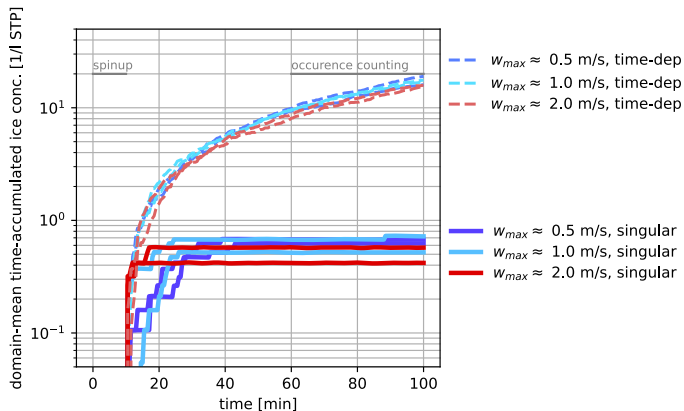
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Testing three flow regimes and two immersion freezing representations

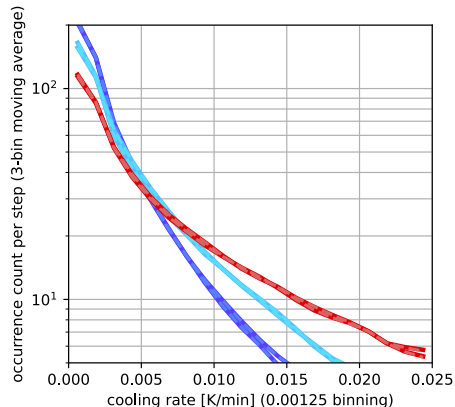
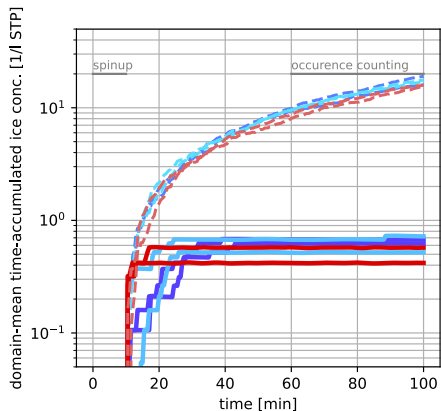


Testing three flow regimes and two immersion freezing representations



► range of cooling rates in simple flow (far from 0.5 K/min for AIDA as in Niemand et al. 2012)

Testing three flow regimes and two immersion freezing representations



- ▶ range of cooling rates in simple flow (far from 0.5 K/min for AIDA as in Niemand et al. 2012)
- ▶ **only time-dependent scheme robust across flow regimes** (consistent with box model & theory)

100%  python™ open-source code:

/ OPEN**ATMOS** /

PySDM

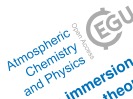
JAMES | Journal of Advances in
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Thank you for your attention!
sylwester.arabas@agh.edu.pl